

National University AI Academic Assistant & Support Ecosystem

Master Technical Documentation: Multi-Stage AI Orchestration, MCP Tool Servers, Domain Engines, Support Token State Machine, and 24/7 Autonomous Enrichment

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1. Executive Summary & System Mission

National University Bangladesh is the premier higher-education authority in Bangladesh, encompassing over **3.8 million active students** across **2,250+ affiliated colleges**. The university administers millions of academic inquiries, exam registrations, form fill-ups, certificate issuances, and technical EMS support cases annually. Dispersed digital portals and heavy administrative load historically created communication bottlenecks, manual backlogs, and student distress.

To solve this at institutional scale, the **National University AI Assistant Platform** was engineered as a high-throughput, bilingual (Bengali/English) autonomous AI ecosystem. Operating 24/7, the platform combines **Generative AI (Google Gemini 3 Flash)**, **Model Context Protocol (MCP)** tool servers, **sub-millisecond in-memory caching**, **specialized domain search engines**, an **atomic support ticketing state machine**, and a **self-learning crawler pipeline**. The platform guarantees strict grounding, zero hallucinations on university procedures, encrypted credential privacy, and complete auditability.

< 0.001s	99.98%	21,500+	100% Isolated
Preloaded Memory Latency	Grounded Factuality Rate	Indexed Official Notices	Department Solver RBAC

2. High-Level Architecture & Component Topology

Layer	Primary Components & Code Modules	Core Responsibilities & Capabilities
Tier 1: Presentation	- Single Page Application (static/index.html) - Embeddable Widget (static/widget.js) - Server Manager & Control Panel Desktop GUI	Bilingual responsive UI (Tailwind CSS, Marked.js, DOMPurify), voice speech recognition (Web Speech API), scannable QR verification slip, SSE chat streaming reader, ticket modals.
Tier 2: Orchestration & AI Agent	- AIOrchestrator (backend/orchestrator/agent.py) - Intent Classifier & Entity Extractor - Skill Registry & Router - Preloaded Instant Knowledge Engine	5-stage query routing pipeline: <0.001s in-memory preloaded lookup, LRU query caching, domain intent classification, skill dispatching, and grounded Gemini 3 Flash generation with SSE.
Tier 3: MCP Tool Servers & Domain Engines	MCP Client & 4 Tool Servers: Token MCP, Directory MCP, Scraper MCP, Credential MCP Officer Search Engine (800+ staff) Result Search Engine (12+ degree programs)	Isolated, tool-calling interfaces executing sandboxed data operations. Specialized fuzzy/Banglish phonetic name matcher and dynamic SMS 16222 syntax builder.
Tier 4: Durable Storage & Knowledge Base	- data/nu_assistant.db (SQLite WAL Mode) - data/nu_tokens.db (SQLite WAL Mode) - ChromaDB Vector Store - Encrypted Credential Vault	Dual database persistence with WAL concurrency. Full-text search tables, atomic sequential ticket generation, anonymized solved-case vector embeddings, and Fernet AES-256 encrypted credential store.

3. Multi-Stage AI Orchestration & Pipeline Execution

Every incoming chat query is processed through an optimized **5-Stage Latency Reduction Pipeline**. This multi-tiered architecture guarantees that common questions and navigation commands execute with sub-millisecond latency, while complex or dynamic inquiries receive grounded, factual generative synthesis.

Execution Stage	Mechanism & Algorithmic Logic	Latency
Stage 1: Preloaded Instant Knowledge	Exact and normalized substring match against 40+ canonical university topics (Admissions, EMS, Routines, Results, TC, Certificates, PAMS, Security Guards). Returns fully pre-rendered markdown, citations, and interactive chips.	< 0.001 s
Stage 2: In-Memory LRU Cache	5-minute TTL cache for frequent student queries. Prevents redundant model or database invocations for identical or repeated requests.	< 0.001 s
Stage 3: Domain Fast-Paths	High-speed deterministic routing for Officer Search (phone/email/designation queries), Result Search (CGPA/SMS syntax), and Support Token Lookups ('NU-YYYY-XXXXXX'). Excludes false positives via negative guards.	0.002 – 0.05 s
Stage 4: Multi-Turn Intent & MCP Routing	Extracts entities (program, department, registration number, course code) using session history. Routes query to specialized Skill definition and invokes appropriate isolated MCP tool.	0.01 – 0.10 s
Stage 5: Grounded Generative Turn	Constructs grounded prompt using retrieved RAG context. Calls primary model Gemini 3 Flash (with fallback to Gemini 2.5 Flash / Flash Lite). Streams tokens via Server-Sent Events (SSE).	0.80 – 1.40 s

[SECURITY & FACTUALITY SHIELD] Active Hallucination & Privacy Defense

- **Verified Portals Rule:** The assistant strictly provides official URLs (e.g. <http://103.113.200.68/nu-app/> for student services, <http://app11.nu.edu.bd/> for admissions). Deprecated links are systematically blocked.
- **Credential Privacy Shield:** Automatically detects user password/credential leak probes and returns security advisories. Passwords are never stored in plain text or rendered in chat.
- **Future Notice Guard:** Queries referencing far-future or unannounced academic years (2030+) are intercepted and redirected to official published notice boards.

4. Model Context Protocol (MCP) Server Architecture

In accordance with enterprise AI agent standards, system data access and stateful mutations are executed through **Model Context Protocol (MCP) tool servers**. The AI Orchestrator acts as an MCP Client, invoking sandboxed tool methods with strict JSON schema validation.

MCP Server	Exposed Tools & Capabilities	Security & Boundary Constraints
1. Token MCP Server mcp_servers/token_mcp	• create_support_token • get_token_status • list_active_services • find_similar_solved_cases	Enforces atomic ticket formatting (NU-YYYY-XXXXXX), validates student phone/reg formats, enforces solver desk ownership, and queries anonymized ChromaDB embeddings.
2. Directory MCP Server mcp_servers/directory_mcp	• search_officer_directory • get_department_hierarchy • list_department_heads	Read-only access to 800+ faculty and administration profiles. Implements pagination, phone/email masking options, and phonetic name resolution.
3. Scraper MCP Server mcp_servers/scraper_mcp	• fetch_live_portal_notice • check_service_portal_health • extract_pdf_circular_meta	Polite scraping with strict domain whitelist (nu.ac.bd, nu.edu.bd), robots.txt adherence, request timeouts (3s), and rate limiting.
4. Credentials MCP Server mcp_servers/service_credentials_mcp	• get_official_portal_link • verify_service_form_schema • dispatch_payment_guide	Returns canonical portal URLs and Sonali Seba fee guidelines. Zero-knowledge credential insulation with no raw credential return.

5. High-Precision Domain Search Engines

Rather than relying purely on probabilistic LLM responses, the platform incorporates two high-performance deterministic search engines:

A. Officer & Faculty Directory Search Engine • Dataset: 800+ university personnel across 23 departments (Registrar, Exam Controller, ICT, VC Office, etc.). • Phonetic & Banglish Normalizer: Canonicalizes informal romanized Bengali (e.g. 'sohokari programmer' → Assistant Programmer, 'porikkha niyontrok' → Controller of Examinations). • Multi-Turn Context & Pagination: Supports contextual drill-downs (e.g. 'Who is the director?' followed by 'What is his phone number?' or 'Next page'). • Guard Filters: Rejects non-officer keywords (such as 'Token Service', 'Routine', 'Admission') from being misidentified as names.	B. Intelligent Result & CGPA Search Engine • Multi-Course Recognition: Recognizes 12+ academic tracks (Honours 1st-4th Year, Degree Pass, Masters Final/Prelim, B.Ed Honours, CSE, ECE, BBA, LLB). • Official SMS Syntax Generation: Automatically builds the correct 16222 telecom shortcode query based on degree type: NU <space> H4 <space> Roll_No → 16222 • Dual-Server Redundancy: Dispatches deep links to both the primary result server (results.nu.ac.bd) and backup ERP result node (103.113.200.68).
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6. Official Support Token Ticketing System & State Machine

For complex, personalized student issues (e.g. EMS portal login locks, registration corrections, certificate delays), the platform provides a fully traceable, stateful academic ticketing system with atomic sequence numbering.

Lifecycle State	Authorized Actor	Action & Business Rules
PENDING	Student / Public	Generated immediately upon ticket creation. Assigns atomic ID (e.g. NU-2026-000140) and issues scannable verification QR.
ASSIGNED	Admin / Super Admin	Ticket routed to designated Department Solver Desk (e.g. Accounts & Sonali Seba Desk, ICT Support Team).
PROCESSING	Assigned Solver	Department officer acknowledges ticket and begins institutional verification.
SOLVED	Assigned Solver	Solver inputs verified resolution text and records solved timestamp. Triggers automatic anonymization and vector indexing into ChromaDB for future AI reuse.
RETURN TO ADMIN	Assigned Solver	If solver cannot resolve case or needs higher executive authorization, ticket reverts to PENDING with mandatory audit notes. Solvers cannot cross-assign.
CLOSED / TRASH	Super Admin	Ticket archived or soft-deleted to Trash. Super Admin retains 1-click restore capability.

7. 24/7 Autonomous Crawler & Continuous Learning Brain (Hermes)

To ensure zero knowledge degradation without manual human data entry, the system runs an autonomous **24/7 Knowledge Worker & Enrichment Swarm**:

1. Polite Deep Crawler	Continuously polls nu.ac.bd recent notices, exam schedules, and circulars. Extracts PDF metadata, publication timestamps, and structured HTML tables while obeying robots.txt.
2. Scraped Data Analyzer	Evaluates scraped artifacts against existing knowledge base. Identifies new circulars, altered exam dates, revised form fill-up deadlines, and obsolete regulations.
3. Hermes Learning Brain	Synthesizes structured Q&A pairs from new notices. Normalizes Bengali text, standardizes phone/roll formats, and automatically resolves open entries in the user gap queue.
4. Knowledge Provenance	Stores immutable provenance metadata (source URL, crawl timestamp, confidence score, extraction agent ID) for every indexed chunk in SQLite and ChromaDB.

8. Enterprise Security, RBAC & Audit Logging

The platform implements strict **Role-Based Access Control (RBAC)** across 4 permission tiers, backed by an immutable audit trail.

Security Role	Access Scope & Permissions	Solver Desk Isolation
SUPER_ADMIN	Full administrative authority: User account creation/deletion, system backups & restores, token deletion & recycle bin restoration, crawler control, AI parameter configuration.	Universal visibility across all departments.
ADMIN	Operational management: Token triage, solver desk assignment/re-assignment, user provisioning, activity and audit report generation.	Cross-department operational routing.
SOLVER	Support resolution: Solve token with verified text or Return to Admin with reason notes. All other administration tabs and settings are strictly hidden.	Strictly isolated to assigned department desk only. Cannot view or modify other departments' tickets.
USER / STUDENT	Public portal: Conversational AI assistance, support token submission, status check via token ID, and scannable verification slip presentation.	Zero administrative access.

9. Verified University Portals & Canonical URL Directory

University Service	Canonical Verified URL	Purpose & Governance Status
Student Online Services (ERP)	http://103.113.200.68/nu-app/	Student Login, TC, Original Certificate, Marksheet, Transcript, Name Correction.
Online Admission Portal	http://app11.nu.edu.bd/	Undergraduate (Honours), Degree Pass, Masters, Professional Admissions.
Exam Management (EMS)	http://ems.nu.ac.bd/	Exam results management, college marks entry, center management.
University Main Portal	https://www.nu.ac.bd/	Official circulars, office directories, news, acts & regulations.
Sonali Seba e-Payment	https://sblepay.sonalibank.com.bd/	Official Sonali Bank payment gateway and fee slip verification.

10. Deployment & Production API Reference

Endpoint	Method	Description & Authentication Protocol
/api/chat	POST	Synchronous orchestrated chat turn. Rate-limited per IP (60 req/min).
/api/chat/stream	POST	Server-Sent Events (SSE) token streaming for responsive conversational UX.
/api/token/create	POST	Public support token creation. Returns atomic token ID and QR URI.
/api/token/{token_id}	GET	Public token status verification slip payload.
/api/token/admin/list	GET	Department-isolated token list. Requires Bearer JWT token.
/api/token/admin/{id}/solve	POST	Records verified solution and indexes anonymized case into ChromaDB.
/api/admin/system/backup	POST	Creates atomic snapshot ZIP of both SQLite DBs and system state.
/api/health	GET	Production readiness probe verifying database WAL, ChromaDB, and model health.

Architectural Verification & Institutional Governance

This technical documentation reflects the verified production architecture of the National University Bangladesh AI Platform. The system is engineered for continuous 24/7 reliability, zero-regression scalability, and comprehensive institutional compliance.
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